Seenu V

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.31
Amalorpavam Higher Secondary School	Class XII (State Board of Tamil Nadu)	2023	99.00%
Amalorpavam Higher Secondary School	Class X (State Board of Tamil Nadu)	2021	93.20%

SCHOLASTIC ACHIEVEMENTS

- **National Talent Search Examination (NTSE) Scholar,** 2021

PROJECTS

Distributed Cryptographic Data Synchronization Engine

Tech Stack: Python, FastAPI, SQLAlchemy, Redis, Celery, PostgreSQL (Supabase), AWS EC2, Docker, GitHub Actions, Ngrok, AES-256

- **Built a containerized backend** on AWS EC2 using Docker Compose. Connected a FastAPI gateway, Redis broker, and Celery workers inside a private virtual network to keep 100% of internal traffic safely hidden from the public internet.
- **Improved webhook performance** by fixing event-loop deadlocks using thread-pooling. Set up exact IPv4 routing through Ngrok, ensuring a 99.9% success rate for receiving incoming messages without server timeouts.
- **Created a fast asynchronous background pipeline** by switching to a local Redis instance. Separated heavy encryption tasks from the main API and removed unnecessary TLS encryption for internal network chatter, cutting response wait times by over 85%.
- **Secured user data** with end-to-end AES-256 encryption processed on private worker nodes. Required strict SSL connections (sslmode=require) to safely store the encrypted data in a remote PostgreSQL database (Supabase).
- **Automated reliable weekly background tasks** using Celery Beat cron jobs. Configured precise UTC time calculations to prevent code-crashing bugs and maintain 100% uptime for scheduled data syncs.
- **Streamlined deployment workflows** by implementing GitHub Actions CI/CD pipelines, automatically testing and pushing new code updates to the AWS server to ensure smooth, error-free production releases.

Automated AI-Powered CI/CD Code Review Pipeline

Tech Stack: Python, Flask, Google Gemini API, PostgreSQL (Supabase), Render, GitHub Webhooks, GitHub REST API.

- **Built a cloud-native webhook server** using Flask to actively listen for GitHub push events, automatically isolating and extracting C++ file changes for real-time algorithmic analysis.
- **Automated code reviews** by integrating the Google Gemini API. Engineered strict prompts to evaluate Big-O Time and Space complexities, generating personalized optimization feedback in under 10 seconds.
- **Tracked developer progression** by setting up a remote PostgreSQL database (Supabase). Queried the last 3 historical reviews per commit to give the AI context, accurately detecting logic regressions over time.
- **Streamlined the CI/CD feedback loop** by utilizing the GitHub REST API. Securely managed access tokens via environment variables to post the AI's final verdict directly back to the commit thread instantly.

Cryptographic Local Credential Vault

Tech Stack: Python, Cryptography (Fernet, PBKDF2HMAC).

- **Engineered a secure credential manager** using Python and the standard Cryptography library. Implemented AES symmetric encryption via the Fernet engine to lock serialized JSON data payloads, ensuring absolute data privacy for local storage.
- **Fortified system authentication** by integrating PBKDF2HMAC key derivation. Generated dynamic 16-byte cryptographic salts (os.urandom) and executed 480,000 hash iterations per save cycle to proactively defend the architecture against brute-force and dictionary attacks.

- **Eliminated volatile memory vulnerabilities** by designing a strict application teardown sequence. Explicitly wiped unencrypted plaintext credentials and keys from RAM upon user exit, neutralizing the risk of credential leakage via memory-dump malware.

SKILLS

Languages: Python, C++, C, SQL

Backend Frameworks & Async Processing: FastAPI, Flask, Celery

Databases & Storage: PostgreSQL, Redis (Upstash)

Security & Cryptography: AES-256 Symmetric Encryption, Fernet, PBKDF2HMAC Key Derivation

Infrastructure, Cloud & CI/CD: AWS (EC2), Docker Containerization, GitHub Actions (Automated CI/CD), PyTest, Git/GitHub, GitHub Webhooks, Render, Supabase, Upstash

Software Engineering & Architecture: Distributed Systems, Asynchronous Execution, Event-Driven Architecture, Microservice Decoupling, LLM API Orchestration, Prompt Engineering

RELEVANT COURSES

- Calculus I & II, Elementary Linear Algebra, Differential Equations.
- Introduction to Programming, Introduction to AI & ML, Numerical Methods, Applied Mathematics.

POSITIONS OF RESPONSIBILITY

Film Director, Behind the Lens:

- Directed films for the institute's filmmaking club, overseeing all aspects of creative production and team coordination.
- Successfully led the team at the Inter-IIT competition for two consecutive years (2024, 2025).